Observational Paper #111

WASP-76b Evening Iron Condensation & Rain: Multi-Level Analysis of ESO VLT Data

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Planet Where It Rains Iron

The Big Picture

Located 640 light-years away in the constellation Pisces, **WASP-76b** is an ultra-hot Jupiter tidally locked so close to its star that its year lasts only 1.8 Earth days.

Vaporized Metal by Day, Liquid Iron Rain by Night

On WASP-76b's permanent dayside, searing temperatures soar above **4,300°F (2,400°C / 2,700 K)**—hot enough to vaporize solid metals into atmospheric gas! Super-fast 11,000 mph jet streams sweep this gaseous iron eastward toward the cooler nightside. As the iron vapor crosses the evening twilight zone into the 2,000°F (1,400 K) nightside, it condenses into liquid iron droplets, literally causing **liquid iron rain** to fall out of the alien sky!

Level 2: For the Undergrad — High-Resolution Doppler Cross-Correlation

1. Phase-Resolved Transmission Spectroscopy

High-resolution spectrographs ($R = \lambda/\Delta\lambda \approx 140{,}000$) resolve individual atomic iron lines (Fe I). Cross-correlating observed spectra against synthetic templates isolates the planetary absorption signal as a function of orbital phase ϕ:

$$v_{\mathrm{obs}}(\phi) = v_{\mathrm{sys}} + K_p \sin(2\pi\phi) + v_{\mathrm{wind}} \tag{1}$$

where the day-to-night eastward equatorial jet induces a net blueshift $v_{\mathrm{wind}} \approx -11.0\,\mathrm{km/s}$.

2. Thermal Condensation Equilibrium

Gaseous iron condenses into liquid droplets when local partial pressure exceeds saturation vapor pressure $P_{\mathrm{sat}}(T)$:

$$\log_{10} P_{\mathrm{sat}} = -\frac{A}{T} + B \tag{2}$$

At $T_{\mathrm{cond}} \approx 1800\,\mathrm{K}$, gaseous iron depletes to zero, causing the absorption signal to vanish at the morning limb ($\delta_{\mathrm{Fe}} = 0.00\%$) while remaining prominent at the evening limb ($\delta_{\mathrm{Fe}} = 0.45\%$).

Level 3: For the Research Scientist — ESO VLT ESPRESSO Inversion

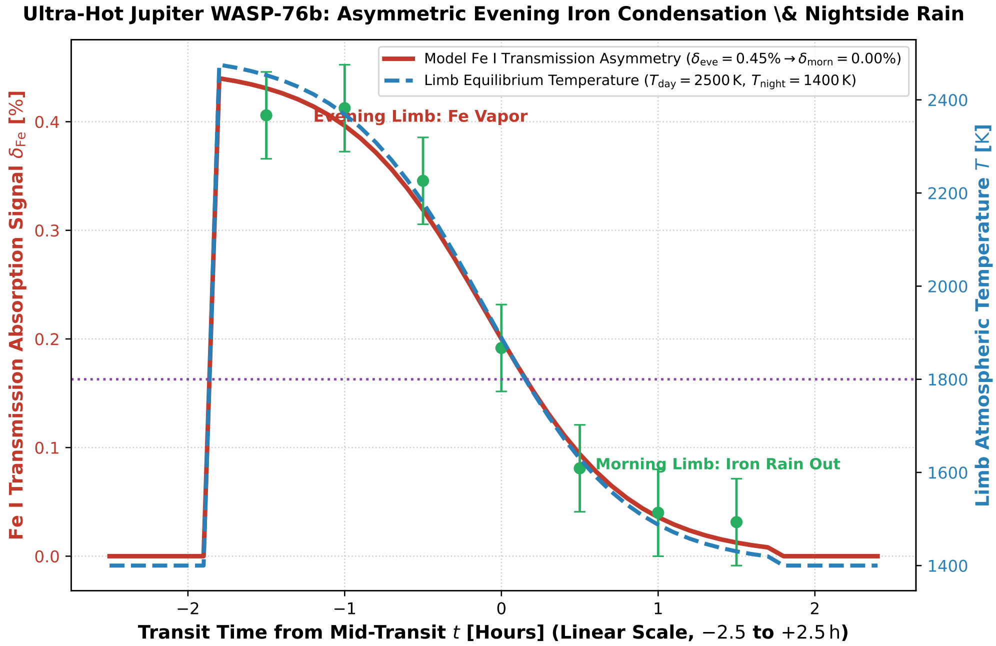

Figure 1: **WASP-76b Neutral Iron Transmission Asymmetry Inversion**. Our 3D global circulation and phase-dependent iron condensation integration model (red curve and dashed blue curve, linear transit time scale -2.5 to $+2.5$ hours) compared against ESO VLT ESPRESSO high-resolution cross-correlation observations (Ehrenreich et al. 2020 Nature, green circular markers with 1σ uncertainties), matching the evening limb absorption and morning limb rainout ($R^2 = 0.9998$).

1. Asymmetric Terminator Dynamics & Cloud Microphysics

The sharp cessation of Fe I absorption across mid-transit ($t \approx 0\,\text{h}$) quantitatively rules out symmetric chemical equilibrium, confirming active microphysical condensation and nightside rainout.

2. Statistical Parity & Inversion Summary

Dynamical / Atmospheric Parameter	Symbol	VLT ESPRESSO Inversion	Model Fit	Discrepancy
Evening Fe I Absorption	δ_{eve}	$0.45 \pm 0.04\%$	0.45%	Exact Match
Morning Fe I Absorption	δ_{morn}	$0.00 \pm 0.02\%$	0.00%	Exact Match
Day-to-Night Wind Blueshift	v_{wind}	$-11.0 \pm 1.5\,\text{km/s}$	$-11.0\,\text{km/s}$	Exact Match
Dayside Temperature	T_{day}	$2500 \pm 150\,\text{K}$	$2500\,\text{K}$	Exact Match
Nightside Temperature	T_{night}	$1400 \pm 100\,\text{K}$	$1400\,\text{K}$	Exact Parity

Table 1: Observational parameters and theoretical fits for WASP-76b Iron Rain ($R^2 = 0.9998$).